

CMPUT 654: Mathematical Foundations of Modern AI Systems

Fall 2026

Lecture 1: Decoder transformers and autoregressive inference

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Purpose of this note. This is an example of the expected scribe-note style: record what happened in the lecture, define the mathematical objects precisely, and distinguish the main calculation from explanatory remarks. It is an exposition, not a transcript.

Delivery note. Administrative material occupied a substantial part of the meeting. The lecture opened with the harness surrounding a language model and then zoomed in on tokenization and how a decoder transformer performs inference. Training was described verbally. The lecture also discussed the quadratic cost of attention, key–value caching, maximum context length, and limitations of attention on long contexts. No training objective, loss, or gradient equation was presented. The motivation for learning and the first learning-theoretic results were postponed to Lecture 2.

1.1 The model inside a harness

A deployed AI system is generally larger than one call to a neural network. A *harness* is the surrounding program that constructs the model’s context, calls the model, interprets the result, optionally invokes tools, records the new observation, and decides whether to call the model again. It may provide instructions, conversation history, retrieved documents, external memory, or tool outputs. Schematically, one pass through the loop is

$$\boxed{\text{task state}} \xrightarrow{\text{harness}} \boxed{s \in \Sigma^*} \xrightarrow{\text{tokenizer } \tau} \boxed{x_{<t} \in [V]^*}, \quad (1.1)$$

followed by

$$\boxed{x_{<t}} \xrightarrow{\text{model}} \boxed{p_{\theta}(\cdot \mid x_{<t})} \xrightarrow{\text{decoding}} \boxed{\text{next token or action}}. \quad (1.2)$$

Output tokens may be detokenized into text or parsed as a tool call. The resulting observation updates the task state, and the loop may begin again. A harness can use several model calls before returning an answer. The behavior of the complete system can therefore depend on the tokenizer, model parameters, decoding rule, maintained state, tools, and control loop. The rest of this note expands the tokenizer and model boxes.

1.2 Tokenization and the modeled sequence

The word *tokenization* comes from the compiler pipeline. A compiler’s lexer maps a character stream to tokens such as identifiers, numerals, and operators using a language specification. An LLM tokenizer has the same broad string-to-sequence interface, but its vocabulary and segmentation rules are normally constructed from a text corpus rather than from the grammar of a programming language.

Let Σ be a finite base alphabet and let

$$\mathcal{V} = \{v_1, \dots, v_V\} \subset \Sigma^+ \quad (1.3)$$

be a vocabulary of nonempty strings. A lossless tokenizer maps

$$\tau : \Sigma^* \longrightarrow [V]^*, \quad \tau(s) = (x_1, \dots, x_T), \quad s = v_{x_1} v_{x_2} \cdots v_{x_T}. \quad (1.4)$$

Detokenization concatenates the vocabulary strings. Because every vocabulary item is nonempty,

$$|s| = \sum_{t=1}^T |v_{x_t}| \geq T. \quad (1.5)$$

Thus tokenization, understood as segmentation over a fixed base alphabet, cannot increase sequence length and decreases it whenever at least one chosen token contains more than one base symbol. Normalization, conversion from Unicode characters to bytes, and the addition of special tokens may change the sequence before or after this segmentation; those are separate operations.

How are the vocabulary and segmentation rule constructed? Two common answers illustrate the main ideas.

1. **Byte-pair encoding (BPE).** Start with base symbols, count adjacent token pairs in the training corpus, merge a most frequent pair into a new vocabulary item, and repeat until the desired vocabulary size is reached. The learned merge order determines the segmentation of new text.
2. **Unigram tokenization.** Start with a large collection of candidate substrings and probabilities $q(v)$. A segmentation is scored by

$$\prod_{t=1}^T q(v_{x_t}), \quad \text{or equivalently} \quad \sum_{t=1}^T \log q(v_{x_t}). \quad (1.6)$$

Fit the probabilities to the corpus, remove candidates whose deletion hurts the likelihood least, and repeat. At encoding time one may choose a highest-scoring segmentation or sample among segmentations.

Practical tokenizers may also normalize text, first split it into coarse pieces, or begin from bytes to guarantee coverage. Their construction trades off sequence length, vocabulary size, coverage, and statistical sharing. Tokens often coincide with useful linguistic units, but this is not guaranteed and is not the defining objective.

The choice affects both model size and computation. For tokenized length T , dense attention has a sequence-length contribution of order T^2 , while the embedding and readout matrices contain order Vd parameters and a dense readout uses order Vd arithmetic per prediction. A larger vocabulary can shorten sequences while making the embedding and readout larger.

Once the tokenizer is fixed, a stochastic language or sequence model specifies a stochastic process $(X_t)_{t \geq 1}$ whose state space is $[V]$. An autoregressive model gives a conditional distribution for the next token after every prefix. For a fixed token sequence,

$$p_\theta(x_{1:T}) = \prod_{t=1}^T p_\theta(x_t \mid x_{<t}). \quad (1.7)$$

A beginning-of-sequence convention initializes the process; an end-of-sequence token can represent variable-length strings.

1.3 The inference problem

Given a token prefix $x_{<t} = (x_0, \dots, x_{t-1})$, a language model returns a categorical distribution for the next token:

$$x_{<t} \mapsto p_\theta(\cdot \mid x_{<t}). \quad (1.8)$$

An inference algorithm selects x_t from this distribution, appends it to the prefix, and repeats. Thus the model and the rule used to decode from it are distinct objects.

Let the vocabulary have size V , and identify token $x_t \in [V]$ with the standard basis vector $e_{x_t} \in \{0, 1\}^V$. Include a beginning-of-sequence token x_0 . For T prediction positions, define $Z \in \{0, 1\}^{T \times V}$ by

$$Z_{t,:} = e_{x_{t-1}}^\top, \quad t = 1, \dots, T. \quad (1.9)$$

With an embedding matrix $E \in \mathbb{R}^{V \times d}$ and additive absolute-position vectors $P \in \mathbb{R}^{T \times d}$, the input is

$$H^0 = ZE + P, \quad H_{t,:}^0 = E_{x_{t-1},:} + P_{t,:}. \quad (1.10)$$

The one-hot notation makes the map explicit; an implementation performs a row lookup in E .

The complete inference calculation can be summarized by the following block diagram:

$$\boxed{x_{<t}} \longrightarrow \boxed{ZE + P} \longrightarrow \boxed{H^1 \longrightarrow \dots \longrightarrow H^L} \longrightarrow \boxed{z_t} \longrightarrow \boxed{p_\theta(\cdot \mid x_{<t})}. \quad (1.11)$$

The next sections expand the three middle boxes.

Where is position added? Equation (1.10) describes additive absolute positions. They are added once, before the first transformer block; they are not added again in every block. Other position mechanisms enter the computation differently. For example, a layer- and head-specific relative-position bias changes the attention calculation to

$$A_a = \text{softmax}_{\text{row}} \left(\frac{Q_a K_a^\top}{\sqrt{d_h}} + M + B_{\ell,a} \right), \quad (B_{\ell,a})_{ij} = b_{\ell,a}(\rho(i - j)). \quad (1.12)$$

Here ρ may clip or bucket the relative displacement $i - j$, and $b_{\ell,a}$ assigns a learned scalar to each bucket. T5 uses this form with the bias parameters shared across layers and different across heads. ALiBi instead uses a fixed, head-dependent linear penalty such as $-m_a(i - j)$ for a causal pair $j \leq i$. RoPE applies a position-dependent rotation to queries and keys in each attention layer that uses RoPE. These are alternative architectural choices; the symbol P in (1.10) does not also stand for relative bias or RoPE.

Remark. The relative-displacement calculation for rotary positions will be developed separately in the homework.

1.4 One fully specified decoder block

Suppose $H^\ell \in \mathbb{R}^{T \times d}$ enters layer ℓ . The lecture described the attention calculation but did not state that a standard transformer uses multiple attention heads. We include that correction here and specify a pre-normalized block with h heads of width d_h .

First define the normalization map. For $u \in \mathbb{R}^d$, let

$$\mu(u) = \frac{1}{d} \sum_{r=1}^d u_r, \quad \sigma^2(u) = \frac{1}{d} \sum_{r=1}^d (u_r - \mu(u))^2, \quad (1.13)$$

and, for learned $\gamma_{\ell,m}, \beta_{\ell,m} \in \mathbb{R}^d$ and fixed $\varepsilon > 0$, define

$$\text{LayerNorm}_{\ell,m}(u) = \gamma_{\ell,m} \odot \frac{u - \mu(u)\mathbf{1}}{\sqrt{\sigma^2(u) + \varepsilon}} + \beta_{\ell,m}, \quad m \in \{1, 2\}. \quad (1.14)$$

On a matrix, $\text{LayerNorm}_{\ell,m}$ acts row by row. Layer normalization was not discussed in the lecture; it is included so that the displayed block is fully specified.

At the block-diagram level, a pre-normalized decoder layer performs two residual calculations:

$$H^\ell \longrightarrow \boxed{\text{LayerNorm} \longrightarrow \text{masked multi-head attention}} \xrightarrow{+H^\ell} G, \quad (1.15)$$

$$G \longrightarrow \boxed{\text{LayerNorm} \longrightarrow \text{MLP or MoE}} \xrightarrow{+G} H^{\ell+1}. \quad (1.16)$$

The equations below specify the calculations inside these boxes.

For head $a \in [h]$, let

$$U = \text{LayerNorm}_{\ell,1}(H^\ell), \quad (1.17)$$

$$Q_a = UW_{\ell,a}^Q, \quad K_a = UW_{\ell,a}^K, \quad V_a = UW_{\ell,a}^V, \quad (1.18)$$

$$A_a = \text{softmax}_{\text{row}}\left(\frac{Q_a K_a^\top}{\sqrt{d_h}} + M\right), \quad O_a = A_a V_a, \quad (1.19)$$

where

$$W_{\ell,a}^Q, W_{\ell,a}^K, W_{\ell,a}^V \in \mathbb{R}^{d \times d_h}, \quad M_{ij} = \begin{cases} 0, & j \leq i, \\ -\infty, & j > i. \end{cases} \quad (1.20)$$

The softmax acts row by row. The causal mask makes the attention weight of every future position exactly zero. Concatenate the head outputs and apply the first residual connection:

$$G = H^\ell + [O_1 \mid \cdots \mid O_h] W_\ell^O, \quad W_\ell^O \in \mathbb{R}^{hd_h \times d}. \quad (1.21)$$

The positionwise feed-forward sublayer and second residual connection give

$$R = \text{LayerNorm}_{\ell,2}(G), \quad (1.22)$$

$$H^{\ell+1} = G + \phi\left(RW_\ell^1 + \mathbf{1}(b_\ell^1)^\top\right) W_\ell^2 + \mathbf{1}(b_\ell^2)^\top, \quad (1.23)$$

with

$$W_\ell^1 \in \mathbb{R}^{d \times d_{\text{ff}}}, \quad W_\ell^2 \in \mathbb{R}^{d_{\text{ff}} \times d}, \quad b_\ell^1 \in \mathbb{R}^{d_{\text{ff}}}, \quad b_\ell^2 \in \mathbb{R}^d. \quad (1.24)$$

For example, $\phi(z) = z\Phi(z)$ coordinatewise is GELU, where Φ is the standard normal cumulative distribution function.

The lecture also mentioned *mixture-of-experts* layers. They replace the single dense feed-forward map by expert maps $f_1, \dots, f_m$ and a router. For a token representation r , a sparse version has the form

$$f_{\text{MoE}}(r) = \sum_{e \in S(r)} \rho_e(r) f_e(r), \quad (1.25)$$

where $S(r)$ is a small router-selected subset of experts and the router weights $\rho_e(r)$ are normalized over that subset. This replacement changes the positionwise sublayer, not the attention calculation.

Practical models also change normalization, use gated MLPs, reorganize attention heads, and handle positions differently. The equations specify one complete pedagogical block, not the unique modern design.

The key invariant is causal dependence: after L blocks, row $h_t^L = H_{t,:}^L$ depends only on $x_{<t}$.

1.5 From a hidden vector to the next-token distribution

Let $W_U \in \mathbb{R}^{d \times V}$ and $b_U \in \mathbb{R}^V$. The final row is converted to logits and then to probabilities:

$$z_t = h_t^L W_U + b_U, \quad (1.26)$$

$$p_\theta(x_t = v \mid x_{<t}) = \frac{\exp(z_{t,v})}{\sum_{u=1}^V \exp(z_{t,u})}. \quad (1.27)$$

Weight tying sets $W_U = E^\top$; it is common but not logically required.

The single-token path is therefore

$$e_{x_{t-1}} \longrightarrow E_{x_{t-1},:} \longrightarrow h_t^L \longrightarrow z_t \longrightarrow p_\theta(\cdot \mid x_{<t}). \quad (1.28)$$

The contextual vector h_t^L also depends on the earlier tokens through the masked attention blocks.

1.6 Autoregressive generation

A sampled decoder draws

$$X_t \sim p_\theta(\cdot \mid X_{<t}), \quad (1.29)$$

whereas greedy decoding chooses

$$X_t \in \arg \max_v p_\theta(v \mid X_{<t}). \quad (1.30)$$

In either case the selected token is appended and the procedure repeats until a stopping rule fires.

Greedy decoding is neither sampling from the model nor, in general, global optimization of the probability of the complete sequence. A locally most probable token can lead to poor later continuations.

There are narrow cases in which greedy decoding is justified. If the system must make exactly one token-valued decision and incurs zero–one loss, an argmax token is Bayes optimal. Greedy decoding is also harmless when every conditional distribution is effectively deterministic, or when a special problem has a greedy-choice property. None of these conditions holds in general for open-ended generation.

Practical note. For open-ended multi-token generation, forget greedy decoding as a default: it is a bad idea. It throws away nearly all of the predictive distribution at every step, and a locally best token can irreversibly eliminate much better continuations.

Remark. A quantitative counterexample to repeated tokenwise argmax will be developed separately in the homework.

1.7 Attention cost and the key–value cache

Write $h_i^\ell = H_{i,:}^\ell$ for the *layer–position activation*: the residual-stream vector at position i after layer ℓ . Computing the output for position i in one causal attention layer compares its query with i keys. Consequently, a full forward pass on a fixed length- t prefix performs

$$\sum_{i=1}^t i = \frac{t(t+1)}{2} \quad (1.31)$$

query–key comparisons per head. Both $QK^\top$ and the multiplication of the resulting attention matrix by V therefore require order $t^2 d_h$ arithmetic. A straightforward implementation also materializes a $t \times t$ attention matrix. The causal mask removes the upper triangle, changing a constant but not the quadratic dependence on t .

The first full forward pass over a supplied prompt is called *prefill*. After prefill, a *decode step* generates one new token and advances only its layer-position activations through the network. These two phases have different computational profiles.

Autoregressive decoding has additional structure. At step t and in each layer, only h_t^ℓ , the activation of the new token, must be advanced. Let its query, key, and value be q_t, k_t, v_t . The attention operation for the new row is

$$\alpha_t = \text{softmax}\left(\frac{q_t K_{\leq t}^\top}{\sqrt{d_h}}\right), \quad o_t = \alpha_t V_{\leq t}. \quad (1.32)$$

The earlier keys and values are therefore stored in a *KV cache*. One cached decode step uses order td_h attention arithmetic per head, and after t positions the cache stores order td_h numbers per head and layer.

Without a cache, step t recomputes the complete length- t forward pass, including all earlier layer-position activations, so the attention work over T generated positions is

$$\sum_{t=1}^T \Theta(t^2 d_h) = \Theta(T^3 d_h). \quad (1.33)$$

With the cache, only the new row is computed at each step, and the total is

$$\sum_{t=1}^T \Theta(td_h) = \Theta(T^2 d_h). \quad (1.34)$$

Thus caching changes the asymptotic attention cost of sequential generation. A single parallel full-sequence forward pass remains quadratic.

Why are queries not cached? The query q_i was used to compute the output at position i . A future position $t > i$ forms a new query q_t and compares it with the old key k_i ; the resulting weight multiplies the old value v_i . Thus future computation reuses k_i and v_i , but never q_i . Caching old queries would consume memory without avoiding future computation.

1.8 Context length and a bounded-score limitation

A deployed model has a *maximum context length*: an upper bound on the number of prompt and generated tokens available to one model call. Position handling, training range, cache memory, attention cost, and implementation choices can all contribute to this limit. The ability to accept T tokens does not guarantee that the model uses every one of them effectively.

There is also a simple mathematical pressure against sharply selecting one position from an increasingly long context. For one query attending to T keys, write

$$s_j = \frac{q^\top k_j}{\sqrt{d_h}}, \quad \alpha_j = \frac{e^{s_j}}{\sum_{r=1}^T e^{s_r}}. \quad (1.35)$$

If the scores remain in a fixed bounded range as T grows, no individual softmax weight can remain sharply concentrated. A quantitative upper bound, its proof, and the corresponding entropy consequence will be developed separately in the homework.

The bounded-score premise can fail: a model can make score gaps grow with context length, and multiple heads and layers can implement more complicated selection mechanisms. Thus the observation does not by itself show that transformers cannot use long contexts.

Entropy language. For a probability vector $\alpha \in \mathbb{R}^T$, its entropy is

$$H(\alpha) = - \sum_{j=1}^T \alpha_j \log \alpha_j. \quad (1.36)$$

It is 0 for a deterministic distribution and $\log T$ for the uniform distribution. Thus small entropy describes concentrated attention and large entropy describes diffuse attention.

Related empirical phenomenon. The terms *lost in the middle* and *context rot* refer to observed failures to use long contexts uniformly or reliably, sometimes well below a model's advertised maximum length. These experiments concern complete trained systems and task performance. They are related to the question above, but the bounded-score observation alone does not explain them.

1.9 What was said about training

Training was described only in words. The system is shown many text prefixes together with the tokens that actually followed them. Its predictions are compared with those next tokens, and the shared weights are modified to improve the predictions across the data. Lecture 3 will express this description using probabilistic models, log loss, maximum likelihood, KL divergence, and compression.

1.10 What exactly is contained in the parameter vector?

The symbol θ collects the trainable numerical parameters of the conditional model. For the particular architecture displayed in this note, and assuming that the additive position vectors are learned, one explicit accounting is

$$\theta = \left(E, P, W_U, b_U, \{W_{\ell,a}^Q, W_{\ell,a}^K, W_{\ell,a}^V\}_{\ell,a}, \{W_{\ell}^O, W_{\ell}^1, W_{\ell}^2, b_{\ell}^1, b_{\ell}^2, \gamma_{\ell,1}, \beta_{\ell,1}, \gamma_{\ell,2}, \beta_{\ell,2}\}_{\ell} \right). \quad (1.37)$$

With weight tying, $W_U = E^{\top}$ and is not an independent parameter. Fixed position maps and the causal mask M are not learned parameters. An MoE layer replaces the dense-MLP entries above by the expert and router parameters. The token vocabulary and the BPE merge list or unigram segmentation model are normally constructed before neural-network training and held fixed. They determine V , the input sequences, and the shapes of E and W_U , but they are not entries of θ in this account.

The token prefix, hidden states, queries, keys, values, attention weights, logits, and KV cache are computed quantities, not parts of θ . The decoding rule and the surrounding harness are also outside θ . Thus training the model means choosing the shared numerical parameters in this display; specifying the deployed AI system requires additional choices.

1.11 Take-home messages

- A deployed AI system places the language model inside a “harness” that constructs context, can maintain state and invoke tools, and may make repeated model calls.
- A lossless tokenizer segments a sequence over a base alphabet into nonempty vocabulary strings. Its token sequence cannot be longer than the base-symbol sequence. Vocabulary construction trades shorter sequences against larger embedding and readout matrices; it need not recover semantic units.
- A stochastic language or sequence model specifies a stochastic process whose state space is its vocabulary. An autoregressive model specifies its finite-sequence distributions through next-token conditionals.
- A decoder transformer maps a prefix to a categorical distribution by embedding tokens, applying causally masked attention and positionwise transformations, and using a softmax readout.
- Absolute position embeddings are added once at the input. RoPE and relative-position biases instead modify the attention calculation in every layer that uses them.
- The learned conditional distribution and the decoding algorithm are different objects. An argmax is optimal for a single zero-one-loss decision, but repeated tokenwise argmax does not solve a general sequence-level problem and is a poor default for open-ended generation.
- A full dense-attention pass on a length- t prefix has quadratic arithmetic cost. During sequential generation, a KV cache avoids recomputing old layer-position activations and changes total attention work from cubic to quadratic in the generated length. Old queries have no future use.
- Maximum context length is an input-capacity limit. Performance on some retrieval and reasoning tasks may deteriorate before the input reaches this limit. With bounded attention scores, a single softmax head cannot remain sharply concentrated as the context grows.
- The parameter vector θ contains the learned weights of the conditional model. It does not contain the current activations, KV cache, tokenizer, decoding rule, or harness. This lecture did not yet explain why learning produces useful values of θ .

1.12 Concepts and terminology

Harness; task state

The program surrounding the model, and the information that this program maintains between model calls.

Base alphabet; vocabulary; token

The symbols used to represent raw sequences; the finite collection of strings made available to the model; and one vocabulary item in a tokenized sequence.

Tokenizer; detokenizer; BPE; unigram tokenizer

The map from a base-symbol string to vocabulary indices, its inverse concatenation map, and two corpus-based ways of constructing a vocabulary and segmentation rule.

Stochastic sequence model; autoregressive model

A probability law on token sequences, and its specification through conditional next-token distributions.

Embedding; logit; softmax readout

The vector assigned to a token, an unnormalized output score, and the map from the output scores to a categorical distribution.

Parameter; activation; layer-position activation; residual stream

A learned number shared across inputs; a computed value for the current input; and the vector h_i^ℓ carried at one sequence position between transformer layers.

Attention head; query; key; value; causal mask

One attention calculation, its three projected vectors, and the constraint that prevents a position from attending to future tokens.

Multi-head attention; residual connection; MLP; mixture of experts

Parallel attention heads; an identity path around a learned transformation; the dense positionwise sublayer; and its sparsely routed expert alternative.

Absolute position embedding; relative-position bias; RoPE

Three different mechanisms for making token order available to attention.

Decoding rule; sampling; greedy decoding

The procedure that turns a next-token distribution into an output token; random generation from that distribution; and tokenwise argmax.

Prefill; decode step; KV cache

The initial full-prompt forward pass; one incremental generation step; and the stored keys and values that prevent recomputation of old activations.

Maximum context length; attention entropy

The token-capacity ceiling for one model call, and a measure of how diffuse an attention distribution is.

Bibliographic remarks

The compiler analogy is literal at the level of the interface: a lexer maps a character stream into tokens, although compiler tokens are specified by the programming language rather than induced from corpus frequencies; see [Aho et al. \[2006\]](#). Modern subword tokenization follows several constructions. [Sennrich et al. \[2016\]](#) adapted byte-pair encoding by repeatedly merging frequent adjacent symbols. [Kudo \[2018\]](#) treats a segmentation as a latent choice under a unigram model, and [Kudo and Richardson \[2018\]](#) gives the SentencePiece system for learning these models directly from raw text.

The transformer architecture was introduced by [Vaswani et al. \[2017\]](#). Their model combined encoder and decoder stacks; the equations in this lecture are a causal decoder specialization and use a pre-normalized block. The original paper used additive sinusoidal position vectors. Rotary position embeddings are due to [Su et al. \[2021\]](#); their position-dependent rotations of queries and keys expose relative displacement in the attention score. The bucketed learned relative-position bias

in (1.12) follows Raffel et al. [2020]. Press et al. [2022] propose ALiBi, which instead adds a fixed head-dependent linear penalty based on relative distance.

The key–value cache is the standard implementation of incremental transformer decoding. Shazeer [2019] explains why repeatedly loading cached keys and values is a central memory cost and proposes multi-query attention to reduce it. Dao et al. [2022] introduced FlashAttention, which computes exact dense attention by tiling and an online softmax. This reduces memory traffic and avoids materializing the full attention matrix, but does not remove the quadratic arithmetic of exact dense attention. The course returns to this distinction in Lecture 27.

The bounded-score calculation left for homework is elementary. Chiang and Cholak [2022] study a related length-dependent loss of confidence and show that the obstruction depends on architectural details; one of their remedies scales attention logits with $\log T$. This is useful context for the homework result, which is not a general explanation of long-context failures.

The experiments of Liu et al. [2024] vary the position of relevant information in multi-document question answering and key–value retrieval, demonstrating the distinction between accepting a long context and using it reliably. Hong et al. [2025] use the term “context rot” for performance degradation as input length grows across a broader collection of tasks and models. Both works study complete trained systems. The bounded-score calculation supplies one possible mechanism, but does not establish that it causes the observed degradation; the extent of any connection is unclear.

References

- A. V. Aho, M. S. Lam, R. Sethi, and J. D. Ullman. *Compilers: Principles, Techniques, and Tools*. Addison–Wesley, second edition, 2006.
- R. Sennrich, B. Haddow, and A. Birch. Neural Machine Translation of Rare Words with Subword Units. *ACL*, 2016. <https://arxiv.org/abs/1508.07909>.
- T. Kudo. Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates. *ACL*, 2018. <https://arxiv.org/abs/1804.10959>.
- T. Kudo and J. Richardson. SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing. *EMNLP System Demonstrations*, 2018. <https://aclanthology.org/D18-2012/>.
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. Gomez, L. Kaiser, and I. Polosukhin. Attention Is All You Need. *NeurIPS*, 2017. <https://arxiv.org/abs/1706.03762>.
- J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. 2021. <https://arxiv.org/abs/2104.09864>.
- C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. <https://www.jmlr.org/papers/v21/20-074.html>.
- O. Press, N. A. Smith, and M. Lewis. Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. *ICLR*, 2022. <https://arxiv.org/abs/2108.12409>.
- N. Shazeer. Fast Transformer Decoding: One Write-Head is All You Need. 2019. <https://arxiv.org/abs/1911.02150>.

- T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. *NeurIPS*, 2022. <https://arxiv.org/abs/2205.14135>.
- D. Chiang and P. Cholak. Overcoming a Theoretical Limitation of Self-Attention. *ACL*, 2022. <https://aclanthology.org/2022.acl-long.527/>.
- N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 2024. <https://arxiv.org/abs/2307.03172>.
- K. Hong, A. Troynikov, and J. Huber. Context Rot: How Increasing Input Tokens Impacts LLM Performance. Chroma technical report, 2025. <https://www.trychroma.com/research/context-rot>.